Queue-aware Scheduling for Shared Research Equipment*

A simulation study of batching and changeover costs

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Abstract

Shared laboratories often schedule instruments in arrival order even when switching between sample families requires substantial setup time. We compare first-come-first-served scheduling with a queue-aware batching rule in a fictional discrete-event simulation. Under moderate demand, batching lowers the share of time spent on changeovers while preserving predictable waiting times. The example also demonstrates optional theorem and table helpers without adding them to the core template.

Keywords: shared facilities, scheduling, batching, discrete-event simulation
JEL Classification: C61, C63

*This demonstration uses invented data and does not describe a real institution or experiment.
†Corresponding author. This paper, its authors, institutions, and results are fictional.

1. Introduction

Universities increasingly operate microscopes, spectrometers, and fabrication tools as shared facilities. A simple arrival-order queue is transparent, but it can trigger repeated changeovers between sample families. When setup consumes a large share of instrument time, a small amount of batching may improve access for the whole user community.

This study compares two scheduling policies in a stylized facility. The first serves jobs in arrival order. The second may hold a job briefly when another job from the current sample family is already waiting. The design goal is not to eliminate waiting, but to reduce avoidable setup while keeping delays bounded.

2. Scheduling model

Let $s > 0$ denote the fixed setup time required when consecutive jobs belong to different sample families. Jobs within a family have identical processing time, and all jobs are available before the schedule begins.

Theorem 2.1. (Single-family batching) For a fixed set of available jobs, processing each sample family in one contiguous batch weakly reduces total setup time relative to any schedule that returns to a previously processed family.

Proof. Every return to a previously processed family creates at least one additional family transition. Merging that family's separated blocks removes a transition and does not change the number or duration of jobs. Repeating this operation produces one block per family with no greater total setup time. □

The proposition motivates batching, but it does not by itself protect individual users from long delays. Our queue-aware policy therefore imposes a maximum hold time before the oldest waiting job must be served.

3. Simulation design

The fictional simulation covers a thirty-day operating period with eight-hour days. Arrivals follow the same demand trace under each policy. We vary only the scheduling rule and summarize one hundred replications.

Policy	Median wait	Setup share	Utilization
Arrival order	46 min	18%	71%
Queue-aware batch	51 min	11%	77%
Long batch	73 min	8%	79%

4. Results

The queue-aware rule reduces the simulated setup share by seven percentage points and increases utilization by six points. Median waiting time rises by five minutes, while the ninety-fifth percentile remains below the configured hold-time limit. The long-batch policy saves more setup time but produces a much larger delay, illustrating why utilization alone is an incomplete objective.

5. Discussion

The results are illustrative, not empirical. Real facilities face cancellations, priority projects, heterogeneous processing times, and maintenance windows. A practical evaluation should preregister service-level outcomes, report effects by user group, and test the policy prospectively before deployment.